

Report on the use of passive acoustic and remote camera monitoring in the Raush Valley

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Figure 1: Site RAU-286B Grizzly Bear

1 Abstract

This report summarizes a pilot biodiversity monitoring program conducted in the Raush Valley watershed using 41 remote cameras and 4 autonomous recording units (ARUs) deployed across 45 sites between 2024 and 2025. The study aims to establish baseline wildlife information, develop standardized data management workflows, and provide recommendations for future long-term monitoring and conservation planning. A total of 29,254 images and 1,093 acoustic recordings were collected and processed through the WildTrax platform using rigorous verification and quality-control procedures. Monitoring documented 38 species, with camera data revealing a diverse mammal community dominated by Hoary Marmot, White-tailed Deer, Moose, and Woodland Caribou, while acoustic surveys detected relatively few notable species due to deployment timing outside peak vocal activity periods. The results highlight the value of passive monitoring technologies for ecological assessment in remote landscapes and provide recommendations for improving future survey design, including earlier-season ARU deployments, standardized recording settings, optimized camera configurations, and improved data organization practices. Together, these findings establish an important ecological baseline for the Raush Valley and support future wildlife management, biodiversity monitoring, and conservation decision-making.

Note

This report is dynamically generated, meaning its results may evolve with the addition of new data or further analyses. For the most recent updates, refer to the publication date and feel free to reach out to the authors.

2 Introduction

Passive environmental monitoring tools, such as trail cameras and autonomous recording units (ARUs), offer a way to observe ecosystems without active human presence. While ARUs (Shonfield and Bayne 2017) are tailored to capture acoustic data from vocal species like amphibians and birds, trail cameras excel at tracking medium to large wildlife. Around the world, conservationists and resource managers are increasingly adopting these technologies because they allow for continuous, long-term monitoring with minimal habitat disruption (Sugai et al. 2018). The insights gathered from these devices can provide critical data to guide ecological management and policy decisions. However, because these sensors make it exceptionally easy to collect large amounts of information, establishing rigorous data integrity standards is essential to guarantee that the data can be easily shared and integrated across different systems in the future.

This project represents a localized subsection of the Fish and Wildlife Compensation Program's grizzly bear population unit, focused within the Raush Valley. Prior to this initiative, there was very little data regarding what wildlife and species actively inhabited the valley. As an initial exploratory project, this baseline monitoring program is intended to evaluate current methods and advise future long-term ecological assessment. While informing grizzly bear management practices remains the primary concern of the study, the data gathered will also play a critical role in guiding future Raush funding and broader ecosystem management strategies.

This report compiles the outcomes of this pilot study to ensure the insights are both transparent and easy for others to replicate. Specifically, this document aims to:

- **Standardize data workflows:** Establish clear, consistent, and reproducible protocols for managing and processing information gathered by remote sensors in the valley.
- **Establish a local wildlife baseline:** Deliver a detailed overview of the various wildlife species identified in the Raush Valley watershed, including their relative presence and activity levels.
- **Guide future strategies and funding:** Utilize preliminary results to guide future funding allocations, track local biodiversity trends, and refine long-term monitoring strategies for the Raush Valley.
- **Inform regional conservation:** Establish an initial ecological foundation to inform grizzly bear management practices and support long-term regional conservation planning.

3 Methods

3.1 Data collection

Cameras and autonomous recording units (ARUs) were deployed across 45 sites in the Raush Valley watershed (Figure 2). This project marked Simpcw Resources Group's first use of ARUs and acoustic monitoring. A total of 45 sensors were deployed, comprising 41 cameras and 4 ARUs, distributed across 17 monitoring blocks in the valley. ARUs recorded for 30 min intervals throughout their deployment until battery failure or SD cards were filled up. Cameras were configured for motion-triggered capture, acquiring a three-image burst per detection event. A total of 1093 recordings and 29254 images were collected, with most data collection spanning between 2024-07-21 and 2025-08-21. Location names were standardized for this report by aligning sites where ARUs and cameras were co-deployed, assigning the ARU records the corresponding camera location name. These changes were applied only within the reporting outputs; original location names in WildTrax remain unchanged.

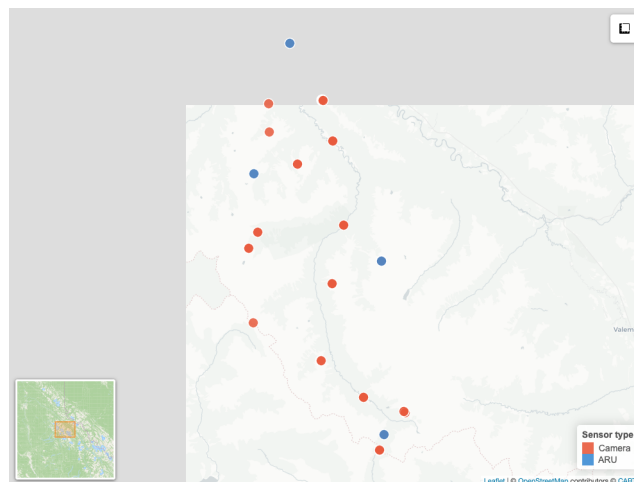


Figure 2: Locations from Simpcw Resources Group ARU and Camera Monitoring Program

Table 1: Locations surveyed.

Show

10

▼
entries

Search:

	Location	Latitude	Longitude	Sensor
1	RAU-286-1A	52.67463	-119.819223	Camera
2	RAU-286-1B	52.67463	-119.819223	Camera
3	RAU-286-1C	52.67463	-119.819223	Camera
4	RAU-286-1D	52.675684	-119.819527	Camera
5	RAU-286-2A	52.695428	-119.810276	Camera
6	RAU-286-2B	52.695598	-119.809932	Camera
7	RAU-296-1A	52.743918	-119.854606	Camera
8	RAU-296-1B	52.743977	-119.854032	Camera
9	RAU-296-1C	52.743922	-119.854058	Camera
10	RAU-297-1A	52.725905	-119.767031	Camera

Showing 1 to 10 of 45 entries

Previous

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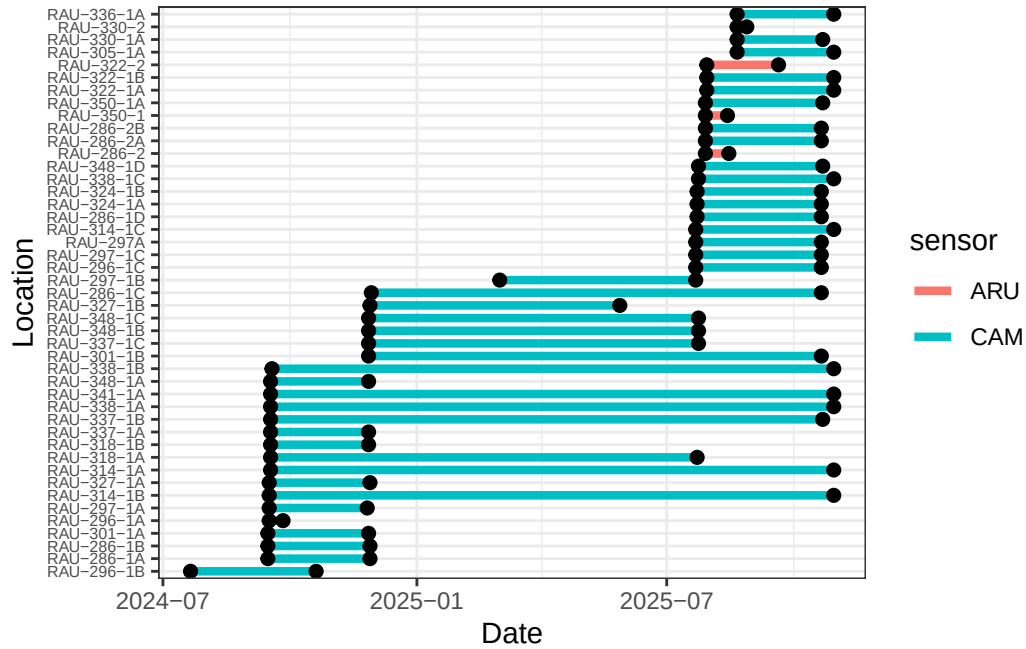


Figure 3: Deployment length of cameras and ARUs.

3.2 Acoustic data

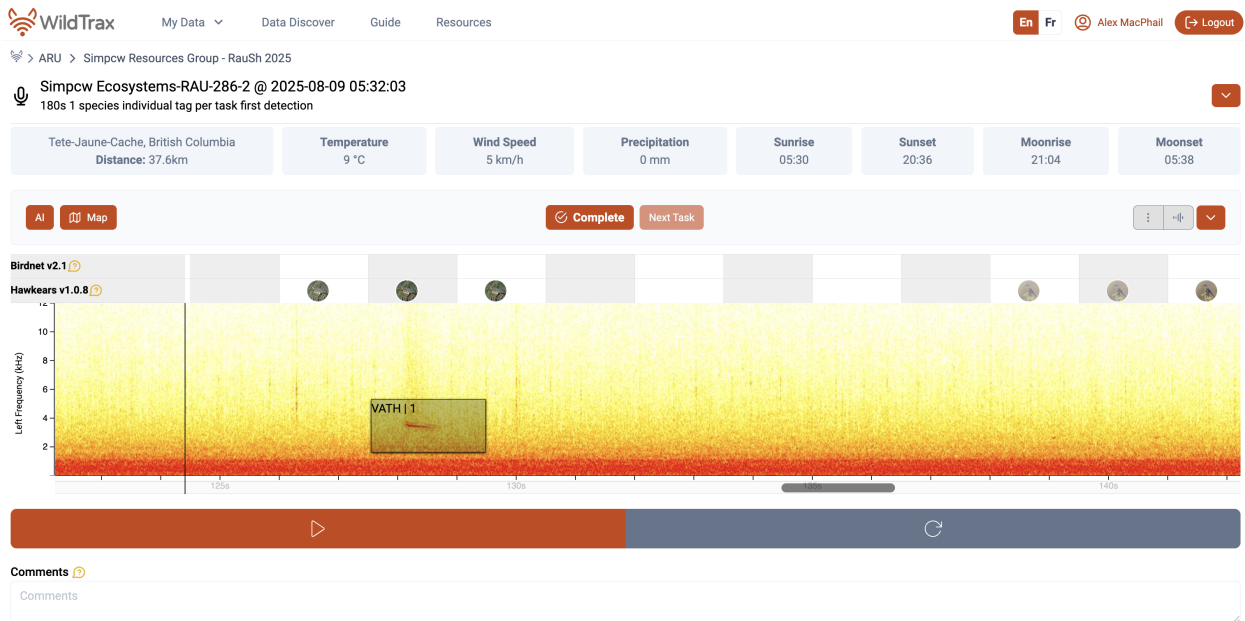


Figure 4: Acoustic tagging interface in WildTrax of a recording from RAU-286-2. Includes a vocalization of a Varied Thrush (VATH).

The principal goal for data processing was to describe the acoustic community of species heard at locations while choosing a large enough subset of recordings for analyses. To ensure balanced

replication, we randomly selected 8 recordings per location and processed each recording for 3 minutes. Four recordings per location were processed between the hours of 4:00 AM - 7:59 AM (dawn period) and four between 20:00 - 23:59 PM (dusk period). Where possible, the 8 recordings from each location were randomly selected from four separate sampling dates. Four samples ensures that there is a minimum number of samples being able to detect most species. Each of the selections were also chosen on days that did not have any inclement weather (heavy rain or wind), which would be unfavourable for bird surveys.

Tags are made using count-removal (Farnsworth et al. (2002), Sólymos et al. (2018)) where tags are only made at the time of first detection of each individual heard on the recordings. Amphibian abundance was estimated at the time of first detection using the [North American Amphibian Monitoring Program](#) with abundance of species being estimated on the scale of “calling intensity index” (CI) of 1 - 3. Mammals such as Red Squirrel, were also noted on the recordings. We also verified that all tags that were created were checked by a second observer to ensure accuracy of detections (Table 2 and Table 3). In addition to human transcription, we also ensured that species were not missed at the location level using additional information from BirdNET (Kahl et al. (2021)) and HawkEars (Huus et al. (2025)) at the task level to avoid false negatives, or missed detections by the human observer. After the data are processed in WildTrax, the `wildrtrax` R package is used to download the data into a standard format prepared for analysis. The `wt_download_report` function downloads the data directly to a R framework for easy manipulation (see [wildrtrax APIs](#)).

3.3 Camera data

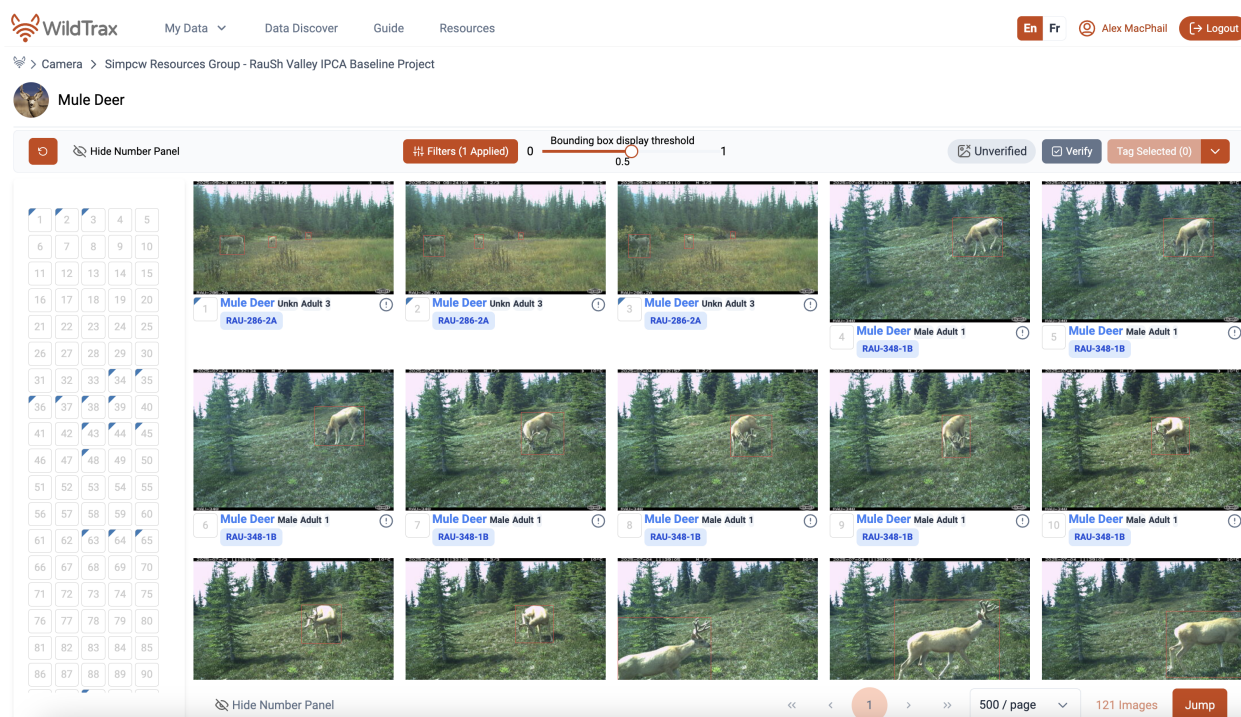


Figure 5: Camera tagging interface in WildTrax. Shows species verification of Mule Deer from the entire Raush dataset.

Table 2: Summary of acoustic verified tags

Show 10 entries

Search:

	Tag is verified	Count	Proportion
1	false	16	42.11
2	true	2	5.26
3		20	52.63

Showing 1 to 3 of 3 entries

Previous

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41 camera deployments were tagged by different observers using the WildTrax platform. Prior to manual tagging, MegaDetector (Beery (2023)) was run on the deployments to auto-tag images with humans, vehicles, and false detections. Bounding boxes were also placed around suspected detections. Detections of animals were tagged for species, and mammal detections received additional tags for count, age, and sex where identification was possible. In addition to animal species, observers manually tagged images with humans, vehicles, and false detections that MegaDetector failed to tag. Tags of mammal species were independently verified by two separate observers who did not participate in initial tagging. Count, age, and sex tags were also verified along with species, and detections tagged as “Unidentified” were verified in case species could be determined. Tags such as human or vehicle, were not verified. Following species verification, tags were quality checked for typos and logical inconsistencies.

4 Results

A total of 38 species were detected across all monitoring sites. Species detections by location are summarized in Table 4, including the maximum count recorded for each species. Shannon diversity (Shannon (1948)) was calculated for each location in the dataset (Figure 6). Shannon diversity is an index that represents both species richness (the number of species present) and evenness (the relative distribution of individuals among those species), providing a single metric of community complexity. Higher values indicate communities with a greater number of species and / or a more even distribution of detections, while lower values reflect communities dominated by fewer species or with uneven detection patterns.

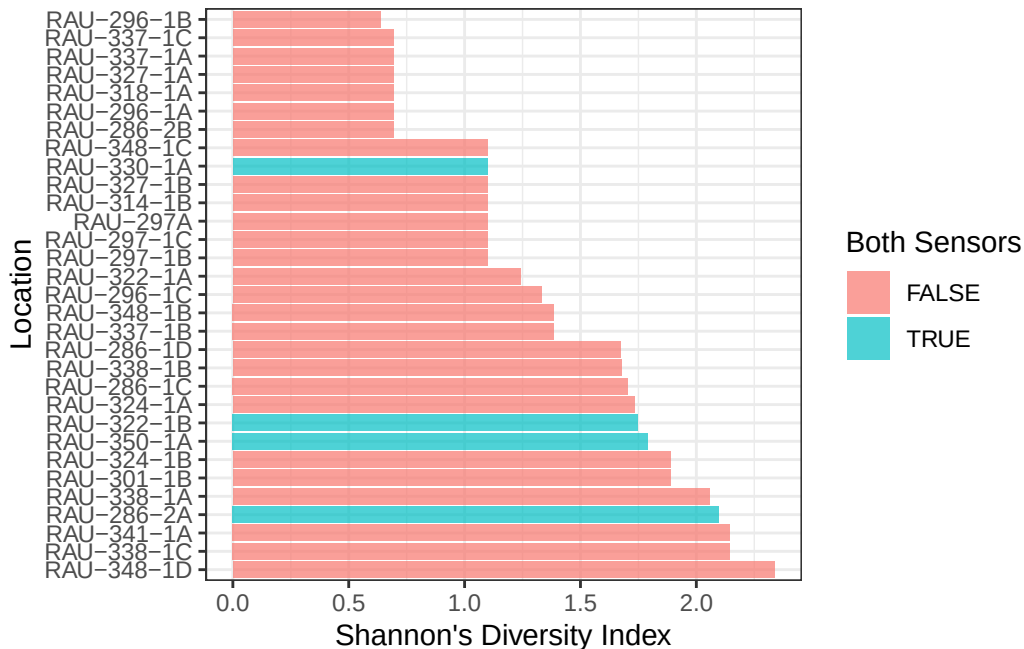


Figure 6: Shannon diversity index at each location. A site with a high Shannon diversity index had many species detected in roughly equal numbers. A site with a low score was dominated by one or two common species, with few others present.

Table 3: Summary of image verified tags

Show entries
Search:

	Tag is verified	Count	Proportion
1	false	24317	86.79
2	true	3702	13.21

Showing 1 to 2 of 2 entries

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Table 4: Summary

Show entriesSearch:

	location	Latitude	Longitude	Sensor	Dark-eyed Junco	Golden-crowned Sparrow	Varied Thrush	Hermit Thrush	Mourning Warbler	Pine Siskin	Woodland Caribou
1	RAU-286-2A	52.695428	-119.810276	ARU	1	1	1				
2	RAU-322-1B	52.919837	-119.815674	ARU				1	1	1	
3	RAU-330-1A	53.032072	-120.089169	ARU						1	
4	RAU-286-1A	52.67463	-119.819223	Camera							1
5	RAU-286-1B	52.67463	-119.819223	Camera							1
6	RAU-286-1C	52.67463	-119.819223	Camera							2
7	RAU-286-1D	52.675684	-119.819527	Camera							3
8	RAU-286-2A	52.695428	-119.810276	Camera							2
9	RAU-286-2B	52.695598	-119.809932	Camera							1
10	RAU-296-1A	52.743918	-119.854606	Camera							

Showing 1 to 10 of 40 entries

Previous

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Next

Camera deployments recorded a diverse assemblage of mammals, with detections dominated by large ungulates and common alpine and subalpine species (see Figure 7). Hoary Marmot was the most frequently detected species (~1,200 images), followed by White-tailed Deer (~750) and Moose (~680). Woodland Caribou were well represented (~620 images) and were detected at all RAU-286 locations, indicating consistent use of the study area. Other notable detections included Grizzly Bear, Mule Deer, and American Pika, though at lower frequencies. Carnivores such as Gray Wolf and Wolverine were detected infrequently. Excluding domestic and non-animal detections (i.e. humans, vehicles, NONE, etc.) 4 camera locations had no motion detection animals images (RAU-297-1A, RAU-314-1C, RAU-318-1B, RAU-348-1A). No notable species detections or species-at-risk were recorded from ARU data, likely reflecting seasonal deployment outside peak vocal activity periods (see Section 5.1).

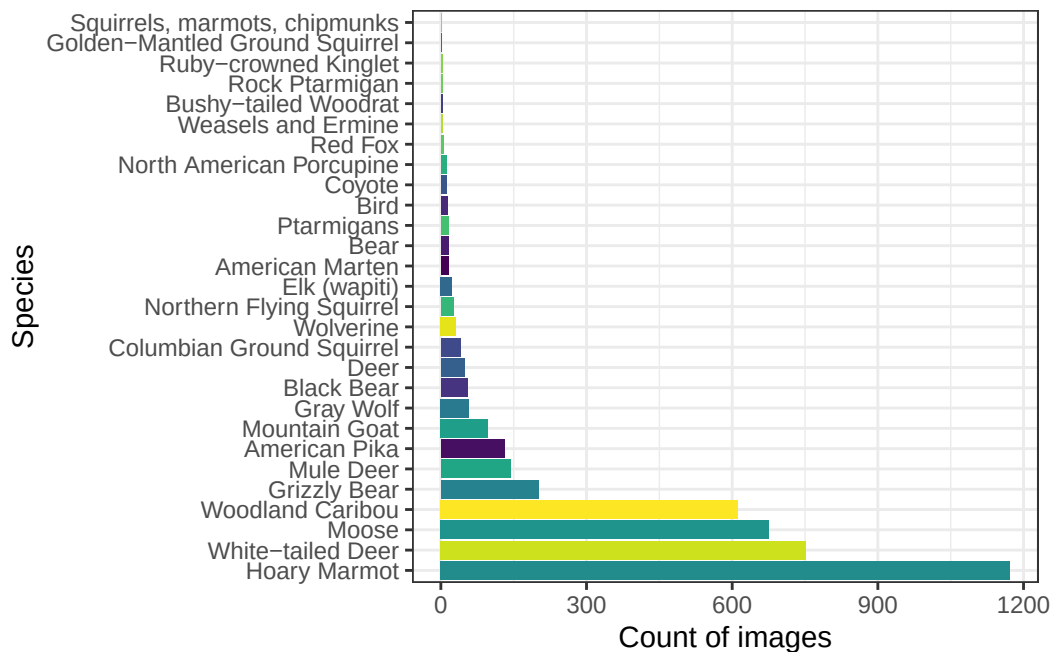


Figure 7: Count of images with species detections.

5 Discussion and recommendations

5.1 Acoustic recommendations

The distribution of ARU recordings reveals two deployment periods concentrated around August 4-11 and August 25, both falling well outside the optimal window for passive acoustic monitoring of birds and amphibians in interior British Columbia. For birds, the primary breeding season, and therefore peak vocal activity, runs from approximately late April through early July. Dawn chorus intensity, which drives the majority of species detections in ARU-based surveys, drops substantially by August as territorial singing ceases following the breeding season. Recordings collected in August are likely to underrepresent species richness and will not be suitable for occupancy or abundance analyses that assume detection probability is tied to breeding activity. For amphibians, calling activity is similarly front-loaded in the season. Most species in the region (e.g., Boreal chorus

frog, Columbia spotted frog) call from snow-melt through June or early July at most. August recordings are unlikely to capture any meaningful calling activity. If site access is constrained by helicopter availability or snow pack, prioritizing early-season access, even for a subset of sites, will yield substantially more useable data than late-summer visits. For future deployments, the following timing is recommended:

- Birds: Deploy ARUs to capture at minimum the period from late April through late June, ensuring coverage of the full dawn chorus window. Ideally, units should be recording continuously or on a duty cycle from deployment through late June.
- Amphibians: Target late April through June, with units placed near wetland and riparian features. Evening and overnight recording schedules are preferable for amphibian detection.

Using a standard recording frequency of 48,000 Hz or 44,100 Hz will align recordings with the conventions used across most bioacoustic workflows, software libraries, and analysis pipelines. The current setting of 48,076 Hz is atypical and can lead to subtle but important issues, including timing drift, incompatibility with resampling routines, and inconsistencies when integrating with reference datasets that assume standard rates. If the same intensive recording schedule is maintained, standardizing the sampling frequency will not materially affect data volume or ecological inference, but will substantially improve interoperability, reproducibility, and processing efficiency.

5.2 Camera recommendations

Despite a high number of unique camera locations and a deployment period of up to one year, the project has a surprisingly low motion image count for a typical project of this size and duration. While a few cameras seem to have been knocked over by animals, such as RAU-296-1A, the primary contributing factor of the low number of images is likely that both images and video were recorded simultaneously. Compared to image files, high definition video files require a large amount of digital storage and quickly fill SD cards, thus preventing further image capture. While video data is effective at short-term monitoring and behaviour capture, image data allows for longer deployment periods and can be more effective at measuring species richness and relative abundance. Focusing on one of these formats in each deployment rather than both will allow for more effective data collection in line with the goals of the study.

Prior to processing, files were structured in a way that required substantial reorganizing before they were ready to upload to WildTrax. For future projects, please ensure that camera data is organized as `Project_Name/Location/Overflow_folders/images.jpg` or `Project_Name/Locations/images.jpg`. Avoid renaming or restructuring the default file structure that your camera provides, as WildTrax is able to recognize these default file structures, but not custom file structures. An example of an optimal ARU and camera file structure for WildTrax uploading has been included on the returning external drive.

Deploying remote cameras and autonomous recording units alongside standardized data workflows establishes a solid baseline for tracking long-term biodiversity trends across the Raush Valley watershed. Refining future survey windows to align with peak seasonal activity will capture more accurate occupancy and richness data for vital avian and mammalian populations. These consolidated datasets will ultimately provide the empirical foundation required to guide local environmental stewardship and protect regional ecological values.

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